

The Imaginary Phase Problem in the Gravitational Path Integral

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Abstract

The sphere partition function is one of the simplest examples for computing the euclidean path integral, usually interpreted as a count of states. However, when we consider the one-loop correction, it contains the dimension dependent imaginary phase, i^{D+2} , which seems confusing for such an interpretation. We show that a consistent state counting interpretation is recovered by requiring only a forward-going clock.

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1 Introduction

The classical action is often referred to as de Sitter entropy, which suggests an interpretation based on state counting [1]. When computing the one-loop correction, a puzzle emerges that Polchinski had observed [2] that the one-loop correction carries a dimension dependent phase

$$\mathcal{Z}[S^D] = \exp\left(\frac{A_c}{4G_N}\right) Z_{\text{one-loop}} \quad , \quad Z_{\text{one-loop}} = i^{D+2} \times (\text{Positive}) \quad (1.1)$$

The imaginary phase originates from divergent integrals in the gravitational path integral, which can be made convergent via contour rotation of the form $c \rightarrow \pm ic$. This approach resembles that used in string theory and the Polyakov path integral, where $X_0 \rightarrow \pm iX_0$. Therefore, the gravitational path integral includes the imaginary phases resulting from this contour rotation.

The presence of this imaginary phase in the gravitational path integral conflicts with the expectation that this quantity should admit a state counting interpretation. To address this problem, Maldacena proposed [3] that including a massive observer cancelled most imaginary phases

$$i^{D+2} \times (-i)^{D-1} = i^3 = -i \quad (1.2)$$

but the suggestion to define the state counting partition function is not well-defined quantity, because an extra minus sign remains

$$Z_{\text{count}} \equiv 2\pi \int dE \rho_{\text{Patch}}(-m - E) e^{S_{\text{clock}}} \frac{m^{D-1}}{(D-1)!} \propto -(\text{Positive}) \quad (1.3)$$

Therefore, there is still some ambiguity in the interpretation of the count of states. Our approach to solving this problem is from a different perspective. As we know, the vacuum state of de Sitter space, $|\Phi_{\text{dS}}\rangle$, can be considered as the euclidean path integral over quantized fields on the hemisphere [4] and also assume that we are in the semi-classical regime, $G_N \rightarrow 0$. So, the inner product of the vacuum state of the de Sitter space becomes equal to the gravitational

path integral of the D -dimensional sphere in the semi-classical limit, which as we have stated, has an imaginary phase. Therefore, if we assume that $|\Phi_{\text{ds}}\rangle$ is a cyclic and separating vector with respect to the von Neumann algebra of type III, then it can be considered the cause of the existence of an imaginary phase that causes the inability to define entropy. In this case, we expect to construct a von Neumann algebra of type II using the structure of crossed product algebra [5], which we can define entropy.

This proposed view is very helpful because we are familiar with the algebraic structure by cross product algebra. On the other hand as we said, the state of the de Sitter space can be expressed as a path integral over quantized fields on the hemisphere, which makes it easy for us to consider new actions. For this reason, we can say that the algebraic correspondence with the path integral formalism allows what is done in the crossed product algebra process to have an effect on the partition function of the system in the path integral formalism. So we expect to be able to define a well-defined quantity of count of states. To understand how to do this, In Section 2, we first review the algebraic structure and Tomita-Takesaki theory. We then examine how a Type III von Neumann algebra can be transformed into a Type II algebra by an additional one-dimensional quantum system to the original system. Then we use a constraint of the generators of each algebra and construct the crossed product algebra, which is an algebra of Type II. So in the algebra of Type II, we can define the trace mapping and well-defined entropy, relative entropy, which agrees with generalized entropy.

Next, in Section 3, we propose a connection between the state of de Sitter space and the euclidean path integral formalism. On the other hand, we assumed the reason of the ill-defined entropy and imaginary phase in the gravitational path integral is that the quantum state of de Sitter space is cyclic and separate with respect to the algebra of Type III. So if we add a one-dimensional quantum system and construct the crossed product algebra, and view the new quantum state as a path integral formalism, then we show if there is a bounded parameter of the second system, q , or namely the "forward-clock going", the correct state counting partition function should defined as

$$Z_{\text{count}} \propto \int_{-i\infty}^{+i\infty} \frac{d\beta}{i}. \quad (1.4)$$

and indeed we have

$$\langle I \rangle \simeq Z_{\text{count}} = \langle \hat{\Phi}_{\text{max}} | \Pi | \hat{\Phi}_{\text{max}} \rangle \quad (1.5)$$

In section 4, according to the correspondence, we show that the existence of the "forward-going clock" or the parameter q in the language of the path integral is a time reversal symmetry breaking. Accordingly, it is natural that the path integral, which is known as the transition amplitude, is in the form of an expression 1.5, and we also conclude that the existence of this symmetry breaking in the de Sitter space is the same problem of correctly defining the correlation function in the semi-classical limit in the de Sitter space, which is mentioned in [6, 7].

2 Modular theory

As is well known in algebraic quantum field theory, rather than directly examining the Hilbert space associated with a subsystem or subregion, one typically studies the algebra of observ-

able operators associated with that region, denoted by $\mathcal{A}$. In most physical examples and phenomena, $\mathcal{A}$ is a von Neumann algebra, classified according to its projector structure into Type I, Type II, and Type III [8]. If we define a map $\omega : \mathcal{A} \rightarrow \mathbb{C}$ called a state, the GNS construction (Gelfand–Naimark–Segal) allows us to reconstruct a Hilbert space from the algebra $\mathcal{A}$. In this representation, there exists a special vector $|\Psi\rangle$ that is cyclic and separating with respect to the algebra $\mathcal{A}$ [9].

Tomita–Takesaki theory states that for any von Neumann algebra $\mathcal{A}$ acting on a Hilbert space $\mathcal{H}$ with a cyclic and separating vector $|\Psi\rangle$, a modular structure is precisely defined, including the following key elements [10, 11]:

- The **modular operator** Δ_Ψ , which leaves the reference vector invariant

$$\Delta_\Psi |\Psi\rangle = |\Psi\rangle. \quad (2.1)$$

- The **modular conjugate** J_Ψ , an anti-linear and densely defined operator that maps the Tomita involution onto the algebra

$$J_\Psi \mathcal{A} J_\Psi = \mathcal{A}', \quad (2.2)$$

where $\mathcal{A}'$ is the commutant of $\mathcal{A}$.

- The **modular flow**, defined as a one-parameter group of automorphisms

$$\sigma_t^\Psi(A) = \Delta_\Psi^{it} A \Delta_\Psi^{-it}, \quad A \in \mathcal{A}, \quad (2.3)$$

known as the modular automorphism group.

This modular structure plays a fundamental role in the connection between algebraic and geometric aspects of quantum field theory. In particular, in relativistic quantum field theories, the modular flow associated with a specific region often corresponds to geometric symmetries of spacetime (such as boost transformations in the Rindler wedge) [12]. The crossed product structure extends this framework by adjoining the modular automorphism group to the original algebra, a new algebra is obtained as [13, 14]

$$\mathcal{A} \rtimes_{\sigma^\Psi} \mathbb{R} \quad (2.4)$$

In other words, adding a one-dimensional quantum system to the original system, the Hilbert space changes as $\mathcal{H} \rightarrow \mathcal{H} \otimes L^2(\mathbb{R})$ and the new algebra can be expressed as

$$\mathcal{A} \rightarrow \mathcal{M} = \mathcal{A} \otimes \mathcal{B}(L^2(\mathbb{R})) \quad (2.5)$$

then if we consider a subalgebra of $\mathcal{M}$ whose elements commute with the sum of the generators of the two new algebra

$$\hat{C} = \hat{K} + \hat{q}. \quad (2.6)$$

this is called a crossed product algebra, denoted by $\widehat{\mathcal{M}}$, and expressed as

$$\widehat{\mathcal{M}} = \{e^{iKp} A e^{-iKp}, e^{-iqs} \mid A \in \mathcal{A}, s \in \mathbb{R}\}'' , \quad (2.7)$$

$$\widehat{\mathcal{M}}' = \{A', e^{-iCs} \mid A' \in \mathcal{A}', s \in \mathbb{R}\}'' . \quad (2.8)$$

where this subalgebra and its complement are von Neumann algebra of type II. More precisely, suppose $\mathcal{A}$ is a Type III algebra, through which we reconstruct the Hilbert space $\mathcal{H}_\Psi^{\text{GNS}}$, where $|\Psi\rangle$ is cyclic and separating with respect to $\mathcal{A}$. Upon adding a one-dimensional quantum system, the new algebra $\mathcal{M} = \mathcal{A} \otimes \mathcal{B}(L^2(\mathbb{R}))$ and the new state can be taken as

$$|\hat{\Psi}\rangle = |\Psi\rangle \otimes |q\rangle \quad , \quad |\Psi\rangle \in \mathcal{H}_\Psi^{\text{GNS}} \quad , \quad |q\rangle \in L^2(\mathbb{R}), \quad (2.9)$$

where $\hat{q}$ and $\hat{p}$ are the position and momentum operators acting on $L^2(\mathbb{R})$, satisfying the usual quantum mechanical commutation relation

$$[\hat{q}, \hat{p}] = i. \quad (2.10)$$

By considering a subalgebra of $\mathcal{M}$ whose elements commute with $\hat{C} = \hat{K} + \hat{q}$, we can form the algebra $\widehat{\mathcal{M}}$. Imposing the constraint $\hat{C}$ means that not every vector $|\hat{\Psi}\rangle$ is physical. Instead, there is a projection operator associated with $\hat{C}$ that defines the physical vector as [15, 16]

$$|\hat{\Psi}\rangle_{\text{P}} = \Pi |\hat{\Psi}\rangle, \quad \Pi = \int_{-\infty}^{+\infty} ds e^{-is\hat{C}}. \quad (2.11)$$

and we can define inner products among physical states as

$$(\hat{\Psi}_{\text{P}}, \hat{\Phi}_{\text{P}}) = \langle \hat{\Psi} | \Pi^2 | \hat{\Phi} \rangle = \langle \hat{\Psi} | \Pi | \hat{\Phi} \rangle. \quad (2.12)$$

Since $\widehat{\mathcal{M}}$ is a Type II algebra [17] so we can define the trace mapping $\text{tr} : \widehat{\mathcal{M}} \rightarrow \mathbb{C}$ as

$$\text{tr}(\rho_{\hat{\Phi}} A) = (\hat{\Phi}_{\text{P}} | A | \hat{\Phi}_{\text{P}}) = \langle \hat{\Phi} | \Pi A | \hat{\Phi} \rangle. \quad (2.13)$$

it can be shown that [9]

$$\rho_{\hat{\Phi}} = 2\pi e^{iK\hat{p}} e^q \Delta_{\Phi\Psi} e^{-iK\hat{p}}, \quad (2.14)$$

where $|\hat{\Phi}\rangle$ is another cyclic and separating physical vector with respect to $\widehat{\mathcal{M}}$. Since our focus is on the constraint $\hat{C}$, we take $\mathcal{A}'$ as the original algebra. In this case, the density operator in 2.14 can be written as

$$\rho'_{\hat{\Phi}} = 2\pi e^{iK\hat{p}} e^q \Delta_{\Phi\Psi}^{-1} e^{-iK\hat{p}}. \quad (2.15)$$

3 dS with an observer

As is well known, apart from black hole horizons, cosmology admits horizons called cosmological horizons. These horizons limit the portion of spacetime that an observer can see or influence. Gibbons and Hawking proposed [1] that in the presence of a cosmic horizon, the quantity $\frac{A}{4G}$, where A is the horizon area, can be interpreted as entropy. The primary motivation was that the area theorem [18] holds equally for black hole and cosmological horizons, showing that A is monotonically non-decreasing. This suggests that $\frac{A}{4G}$ admits a thermodynamic interpretation for cosmological horizons, just as for black holes.

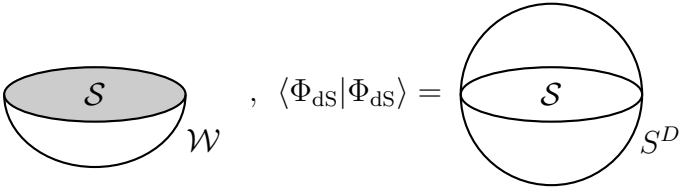
Now, considering the de Sitter space, we first examine the algebraic structure and the effect of adding a one-dimensional quantum system to de Sitter space on the path integral formalism.

3.1 The state counting partition function

In quantum field theory on a de Sitter background, it is natural to take the invariant state $|\Phi_{\text{ds}}\rangle$ as the Bunch-Davies vacuum. This state is defined by analytic continuation to the Euclidean signature (see Appendix A). As we know, the path integral over the quantized fields on the hemisphere $\mathcal{W}$ can be viewed as preparing a quantum state in de Sitter space, additionally we consider this system in semi-classical regime, $G_N \rightarrow 0$. Therefore

$$|\Phi_{\text{ds}}\rangle = \frac{1}{\sqrt{Z}} |\Psi_{\mathcal{W}}\rangle = \frac{1}{\sqrt{Z}} \int_{\phi_{\mathcal{W}}|_{\mathcal{S}}=\phi_{\mathcal{S}}} Dg D\phi_{\mathcal{W}} e^{-I[g,\phi_{\mathcal{W}}]} \quad , \quad G_N \rightarrow 0 \quad (3.1)$$

where $I[g, \phi_{\mathcal{W}}]$ is a Einstein-Hilbert gravity action. So we can consider this quantum state corresponds to a hemisphere and the inner product of this state with itself corresponds to gluing two hemispheres together and equal to the one-loop gravitational path integral (without normalization)

$$|\Phi_{\text{ds}}\rangle = \text{hemisphere } \mathcal{S} \text{ on } \mathcal{W} \quad , \quad \langle \Phi_{\text{ds}} | \Phi_{\text{ds}} \rangle = \text{glued sphere } \mathcal{S} \text{ on } S^D = \mathcal{Z}[S^D]$$


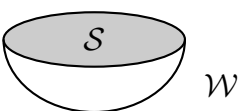
This perspective is very useful because as we know, the sphere partition function contains an imaginary phase [2]

$$\mathcal{Z}[S^D] = \exp\left(\frac{A_c}{4G_N}\right) Z_{\text{one-loop}} = i^{D+2} e^{S_{\text{ds}}} \times (\text{Positive}) \quad (3.2)$$

so the imaginary phase that causes ambiguity in the interpretation of the count of states can be seen as a consequence of the $|\Phi_{\text{ds}}\rangle$ being a cyclic and separating vector with respect to the algebra of type III. It is therefore natural that we cannot interpret the entropy in terms of state counting

$$\text{tr}(\mathbf{I}) \neq Z = \langle \Phi_{\text{ds}} | \Phi_{\text{ds}} \rangle \quad (3.3)$$

We expect that the solution lies in converting this algebra to a Type II algebra, which, as discussed in the previous section, requires adding a one-dimensional quantum system to the original system. Therefore, we start by writing

$$|\Phi_{\text{ds}}\rangle = \frac{1}{\sqrt{Z}} \int_{\phi_{\mathcal{W}}|_{\mathcal{S}}=\phi_{\mathcal{S}}} Dg D\phi_{\mathcal{W}} e^{-I[g,\phi_{\mathcal{W}}]} = \text{hemisphere } \mathcal{S} \text{ on } \mathcal{W} \quad , \quad G_N \rightarrow 0$$


Therefore, we represent the algebra in the de Sitter space by $\mathcal{A}$, where $|\Phi_{\text{ds}}\rangle$ is the cyclic and separating vector with respect to this algebra, and K is also expressed as the Hamiltonian (generator) operator. So by adding a one-dimensional quantum system to the original system which q and p are position and momentum operator, respectively. Then the new algebra, $\mathcal{M} = \mathcal{A} \otimes L^2(\mathbb{R})$, and the new state can be written as $|\hat{\Phi}\rangle = |\Phi_{\text{ds}}\rangle \otimes |q\rangle$. In this case, $\mathcal{M}$ is still a type III algebra, even with the addition of a new degree of freedom. Therefore we consider the

constraint

$$\hat{C} = K + q = 0 \quad (3.4)$$

Therefore considering the constraint, we can consider a subalgebra of $\mathcal{M}$ such that all its elements are commutent by $\hat{C}$, and we call it a crossed product algebra and denote it by $\widehat{\mathcal{M}}$. As mentioned earlier, physical states are defined as $|\hat{\Phi}\rangle_{\text{P}} = \Pi |\hat{\Phi}\rangle$ and also by the definition of the inner product as 2.12, we can define the trace

$$\text{tr}(\rho_{\hat{\Phi}} A) = (\hat{\Phi}_{\text{P}} | A | \hat{\Phi}_{\text{P}}) = \langle \hat{\Phi} | \Pi A | \hat{\Phi} \rangle \quad (3.5)$$

where

$$\rho_{\hat{\Phi}} = 2\pi e^{iK\hat{p}} e^q \Delta_{\Phi\Psi}^{-1} e^{-iK\hat{p}} \quad (3.6)$$

Since all cyclic and separating vectors for the algebra $\mathcal{A}$ in de Sitter space are equivalent, so $\Delta_{\Phi\Psi} = \Delta_{\Phi} = e^{-K\hat{\Phi}}$. Therefore if we taking $A = e^{-q-K} \in \widehat{\mathcal{M}}$ and redefine $|\hat{\Phi}\rangle \equiv \sqrt{\beta_{\text{ds}}} e^{-\frac{\beta_{\text{ds}}}{2} q} |\hat{\Phi}\rangle^1$ we can obtain

$$\text{tr}(\rho_{\hat{\Phi}} e^{-q-K}) = \text{tr}(2\pi e^q \Delta_{\Phi\Psi}^{-1} e^{-q-K}) = \text{tr}(2\pi \text{I}) = (\hat{\Phi}_{\text{P}} | e^{-K} | \hat{\Phi}_{\text{P}}) \equiv \langle \hat{\Phi} | \Pi e^{-K} | \hat{\Phi} \rangle \quad (3.8)$$

where $|\hat{\Phi}\rangle$ is as maximally mixed state $|\hat{\Phi}\rangle \equiv |\Phi_{\text{max}}\rangle = \sqrt{\beta_{\text{ds}}} |\Phi_{\text{ds}}\rangle e^{-\frac{\beta_{\text{ds}}}{2} q} |q\rangle$ Therefore, the physical state is written as follows

$$|\hat{\Phi}\rangle_{\text{P}} = \Pi |\Phi_{\text{max}}\rangle = \Pi e^{-\frac{K}{2}} |\Phi_{\text{ds}}\rangle \otimes \int dq e^{-\frac{q}{2}} |q\rangle \quad (3.9)$$

then

$$2\pi \text{tr}(\text{I}) = (\hat{\Phi}_{\text{P}}, \hat{\Phi}_{\text{P}}) = \langle \Phi_{\text{max}} | \Pi | \Phi_{\text{max}} \rangle \quad (3.10)$$

On the other hand, if we rewrite equation 3.1 in the energy basis, we have

$$2\pi \text{tr}(\text{I}) = -\frac{i}{Z} \int_{-i\infty+\beta_{\text{ds}}}^{+i\infty+\beta_{\text{ds}}} d\beta \int dE \rho(E) e^{-\beta E} \int dq e^{-\beta q} \quad (3.11)$$

But the above expression is still incomplete. As we know from crossed product algebra, the parameter q , which is the position operator of a one-dimensional quantum system, if it is in the $(-\infty, +\infty)$, then $\text{tr}(\text{I}) = \infty$ and the above expression is meaningless. Therefore, if we consider q to be bounded, $q \geq 0$ then $\text{tr}(\text{I}) = 1$. Thus

$$Z = \int_{-i\infty+\beta_{\text{ds}}}^{+i\infty+\beta_{\text{ds}}} \frac{d\beta}{2\pi i} \int dE \rho(E) e^{-\beta E} \int dq e^{-\beta q} \quad (3.12)$$

From the algebraic structure, the equation 3.12 represents a new quantity. However, this

¹As we know the inner product of the former state was

$$\langle \hat{\Phi} | \hat{\Phi} \rangle = \int dq e^{-q} \langle \Phi_{\text{ds}} | \Phi_{\text{ds}} \rangle \quad (3.7)$$

so we just define the new state and introduce β_{ds} as a new parameter to normalized state.

expression is not yet complete, since we have no information about the properties of this expression except that it is a normalization term and in a special case, is equal to the gravitational path integral of the sphere. But we cannot conclude with certainty that after applying the crossed product algebraic structure, this quantity is still a physical quantity and partition function or can be any arbitrary mathematical quantity² Therefore, what resolves this ambiguity is to consider the physical language, or path integral formalism, and because of the proposal we presented in 3.1 we have a very easy task. Namely, we can translate what is a coincidence in the algebraic perspective into the path integral formalism.

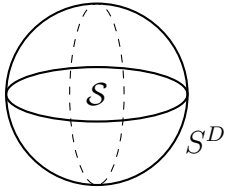
In this framework, the first integral naturally corresponds to the Euclidean path integral over the hemisphere, while the second integral, associated with the one-dimensional quantum system, can be interpreted as an additional path integral governed by an effective action, which we refer to as the observer action. We consider the toy model action for the one-dimensional quantum system, on which the observer moves on the geodesic [19, 20], so the action is $I[q] = \oint_{\gamma} d\tau \left(p \frac{dq}{d\tau} - q \right) = -\beta_{\text{ds}} q$. For the operator Π , because in the algebraic structure it appears as a constraint so we can argue in the path integral formalism it can be as a Lagrange multiplier enforcing the corresponding constraint. So we have

$$\begin{aligned} \Pi &\rightarrow \int D\lambda \exp(+\lambda C) = \pm 2\pi i \delta(C), \\ \int dq e^{-q/2} |q\rangle &\rightarrow \int Dq e^{-I[q]} = \int dq e^{-\beta_{\text{ds}} q}. \end{aligned} \quad (3.13)$$

This yields the path integral expression for the physical state

$$|\hat{\Phi}\rangle_{\text{P}} = \frac{1}{\sqrt{Z}} \int_{\phi_{\mathcal{W}}|_{\mathcal{S}}=\phi_{\mathcal{S}}} Dg D\phi_{\mathcal{W}} Dq D\lambda e^{-I[g,\phi_{\mathcal{W}}]-I[q]+\lambda C}. \quad (3.14)$$

so can be shown the resulting geometric inner product reproduces the result of [20, 21] for the partition function in the presence of an observer.

$$\text{tr}(\mathbf{I}) = (\hat{\Phi}_{\text{P}}, \hat{\Phi}_{\text{P}}) = \frac{1}{Z} \int_0^{+\infty} dE e^{S_{\text{ds}} - 2\beta_{\text{ds}} E} = \frac{1}{Z} \frac{e^{S_{\text{ds}}}}{2\beta_{\text{ds}}} = \frac{\text{Sphere}}{S^D} = 1.$$


therefore the partition function of the physical state in the path integral formalism is

$$Z = \exp(S_{\text{ds}}) = Z_{\text{count}}. \quad (3.15)$$

Thus, we can conclude that the equation 3.12 obtained from an algebraic point of view is a physical quantity which is the state counting parsing function. So in contrast to [3], the correct state counting partition function for the sphere including an observer is

$$Z_{\text{count}} = \exp(S_{\text{ds}}) = \int_{-i\infty+1}^{+i\infty+1} \frac{d\beta}{2\pi i} \int dE \rho(E) e^{-\beta E} \int dq e^{-\beta q}. \quad (3.16)$$

²Even if it is a physical quantity, it can still have an imaginary phase. In that case, the conclusion we draw is that there will still be ambiguity in the interpretation of entropy.

Furthermore, if we add a constant m representing the observer's mass, the state counting partition function is

$$Z_{\text{count}} = \exp(S_{\text{dS}} - \beta_{\text{dS}} m) = \int_{-i\infty+\beta_{\text{dS}}}^{+i\infty+\beta_{\text{dS}}} \frac{d\beta}{2\pi i} \int dE \rho(E) e^{-\beta E} \int dq e^{-\beta q} e^{-\beta m}. \quad (3.17)$$

3.2 On the overall phase

As review the problem, the gravitational path integral for the sphere contains imaginary phase in the semiclassical limit, which is not a well-defined quantity to interpret as a count of states. As noted in [3], the idea was to add a massive observer [22] with a clock and then obtain the partition function of a sphere that includes an observer. But it is still not a well-defined physical quantity because the state counting partition function has an extra minus sign

$$Z_{\text{count}} = \int_0^{+\infty} \frac{d\beta}{i} \int dE \rho(E) e^{-\beta E} \int dq e^{-\beta q} e^{-\beta m} = -i \times (\text{Positive}) \quad (3.18)$$

So we suggest solving this extra minus from a different perspective by considering the partition function of the sphere is equal to the norm of the $|\Phi_{\text{dS}}\rangle$ and this state is cyclic and separating with respect to a Type III von Neumann algebra, therefore the entropy cannot be properly defined. Namely, because in the algebra of Type III the trace mapping does not exist, we have

$$\text{tr}(\mathbf{I}) \neq Z = \langle \Phi_{\text{dS}} | \Phi_{\text{dS}} \rangle. \quad (3.19)$$

Therefore, if adding a one-dimensional quantum system to the original system and assuming $q \geq 0$ then we can obtain a Type II₁ von Neumann algebra and making entropy well-defined. In this approach we obtain the correct state counting partition function

$$Z_{\text{count}} = \int_{-i\infty+\beta_{\text{dS}}}^{+i\infty+\beta_{\text{dS}}} \frac{d\beta}{2\pi i} \int dE \rho(E) e^{-\beta E} \int dq e^{-\beta q} e^{-\beta m} \sim 1 \quad (3.20)$$

which is consistent with the result of the state counting partition function in [23], this partition function has no extra minus sign. So the notable feature is that a well-defined state counting partition function emerges from the modified maximally mixed state $|\hat{\Phi}_{\text{max}}\rangle$, by introduce the operator Π . Namely by considering $|\Phi_{\text{dS}}\rangle$, the partition function of the system has an ill-defined interpretation of the count of states, even if we add a new degree of freedom alone as q and the state of de Sitter space transforms to a maximally mixed state

$$\text{tr}(\mathbf{I}) \neq Z_{\text{count}} = \langle \Phi_{\text{max}} | \Phi_{\text{max}} \rangle \quad (3.21)$$

However, the interpretation remains correct when we use Π in 3.10, which means we should consider the modified maximally mixed state to remove the overall phase

$$\text{tr}(\mathbf{I}) = Z_{\text{count}} = \langle \Phi_{\text{max}} | \Pi | \Phi_{\text{max}} \rangle = \int_{-i\infty+\beta_{\text{dS}}}^{+i\infty+\beta_{\text{dS}}} \frac{d\beta}{2\pi i} \int dE \rho(E) e^{-\beta E} \int dq e^{-\beta q} e^{-\beta m} \quad (3.22)$$

4 A Theory with broken time reversal symmetry

In quantum gravity with Euclidean signature, the scattering amplitude is generally given by the bilinear form $(\cdot, \cdot)$. This implies that for our geometry $Y = X \times I$, the two boundaries are indistinguishable, and swapping them has no effect on the calculation.

However, when working with the quantum mechanical bracket $\langle \cdot | \cdot \rangle$, the two forms are not always equivalent. For example, in theories with a θ -term, the orientation on the manifold prevents a simple identification of the two inner products. It has been shown that [24]

$$\begin{cases} Z = (\psi, \phi) = \langle \psi | \phi \rangle & \text{if reflection symmetries are preserved,} \\ Z = (\psi, \phi) = \langle \psi | T | \phi \rangle \neq \langle \psi | \phi \rangle & \text{if reflection symmetries are broken,} \end{cases}$$

where T is an antiunitary operator. As we have emphasized, we were able to reconstruct equation 3.10 geometrically only by providing a path integral representation for each term in equation 3.9. This suggests that we are dealing with a euclidean action that includes a Lagrange multiplier and describes an observer moving along a geodesic. Then the effective action can be written as

$$\begin{aligned} -I_E &= -\frac{1}{16\pi G_N} \int d^D x \sqrt{g} (R - 2\Lambda) + \oint d\tau \left[ip(\tau) \frac{dq(\tau)}{d\tau} - q(\tau) \right] \\ &= -\frac{1}{16\pi G_N} \int d^D x \sqrt{g} (R - 2\Lambda) - \oint d\tau q(\tau). \end{aligned} \quad (4.1)$$

Now given that q is the Hamiltonian of the observer's clock and p is the time of the clock, so we can show if the Hamiltonian constraint is valid then $p = \tau$ [20]. Therefore, by imposing a bound on the parameter ($q \geq 0$), the time of the clock has a direction and explicitly break reflection symmetry or more precisely, time-reversal symmetry. This is precisely why the partition function in equation 3.22 takes the form

$$Z_{\text{count}} = (\hat{\Phi}_P, \hat{\Phi}_P) = \langle \Phi_{\text{max}} | \Pi | \Phi_{\text{max}} \rangle. \quad (4.2)$$

Thus introducing a bounded parameter as $q \geq 0$ related to the definition Π , and because this boundedness means time reversal symmetry breaking. So we can argue that time reversal breaking is equivalent to defining Π and is an important point to have a well-defined state counting partition function.

On the other hand, in [6, 7], the issue addressed by the author concerns the proper definition of correlation functions in the semiclassical limit in de Sitter space. The author argues that, in order to define this quantity consistently, one must introduce a physical object namely, a clock which is required to be "forward-going" or in other words has a direction. More precisely, let $A(t)$ be a Hermitian operator in de Sitter space and consider the quantity $C[A(t), A(-t)]$. One can then show that, in the presence of the "forward-going clock"

$$\langle C \rangle = \text{tr}(C) = \langle \Phi_{\text{max}} | C \Pi | \Phi_{\text{max}} \rangle \quad (4.3)$$

As can be seen, this expression is closely related to 4.2. Namely

$$\langle I \rangle = \text{tr}(I) = \langle \Phi_{\text{max}} | \Pi | \Phi_{\text{max}} \rangle \simeq Z_{\text{count}} \quad (4.4)$$

Based on this, the resolution of both issues relies on the presence of symmetry breaking through the introduction of a directed object. Furthermore, an additional point is that, in the problem of the imaginary phase in the gravitational path integral, an instability in the path integral gives rise to the imaginary contribution, and by selecting a preferred direction, this phase is removed. In the case of the correlation function, the de Sitter vacuum state is itself unstable [6, 7]. Accordingly, by choosing a direction this instability is converted into an asymmetric but stable configuration. So, by identifying the de Sitter vacuum with the gravitational path integral in the limit $G_N \rightarrow 0$, the problem of the imaginary phase becomes equivalent to the problem of defining correlation functions in the semiclassical limit.

Therefore, because the state counting partition function is obtained from the correct definition of the correlation function in the semi-classical limit, we can conclude that what is proposed in 3.1 can be claimed to be logical.

5 Conclusion

We have shown that the path integral over the hemisphere can be interpreted as preparing a quantum state in de Sitter space in the limit $G_N \rightarrow 0$, so the inner product of this state is the sphere partition function, which contains the imaginary phase and there is no well-defined interpretation as a count of states. So we can assume the quantum state is cyclic and separating with respect to a Type III von Neumann algebra. Therefore we expect that converting the algebra to an algebra of Type II will allow a proper definition of the state counting partition function. This allows us to define the mapping tr and by bounding the Hamiltonian of the new system q , we have $\text{tr}(\mathbf{I}) = 1$ which we can obtain equation 3.12.

Now, this quantity alone does not give us any specific physical meaning and we need to use the gravitational path integral. Therefore, according to the proposal, what we used in the algebraic perspective of the crossed product algebra structure process can be viewed in the more physical language of the path integral. So view this state in the path integral formalism, where the inner product of this state is the state counting partition function. It means the state counting partition function differs from the proposal in [3], which carries an extra minus sign

$$Z_{\text{count}} \propto -i \int_0^\infty d\beta, \quad (5.1)$$

and obtain the correct form

$$Z_{\text{count}} \propto \int_{-i\infty}^{+i\infty} \frac{d\beta}{i}. \quad (5.2)$$

where the imaginary phase in the gravitational path integral is removed. Therefore the massive or massless observer is not the main point to cancel the imaginary phase and it just requires only a bounded parameter in the one-dimensional quantum system, which means $q \geq 0$. On the other hand we conclude the correct form of the state counting partition function needs the modified maximally mixed state

$$Z_{\text{count}} = \langle \Phi_{\text{max}} | \Pi | \Phi_{\text{max}} \rangle \quad (5.3)$$

Furthermore, we showed that the state counting partition function is based on the breaking

of time reversal symmetry, which is therefore equivalent to the correct form of the correlation function in the semi-classical limit in de Sitter space. We can also respond to the paper [7] by considering the instability of spontaneous breaking of time reversal symmetry (SSB) as equivalent to the Polchinski instability, which leads us to the conclusion that the two-point correlation function in the semi-classical limit in the de Sitter space becomes the same as the state counting partition function in this space.

Therefore, it is possible that our proposed correspondence is a physical one due to the correct form of the partition function and the alignment with the correlation function in the semi-classical limit in de Sitter space, but it is still unclear why this correspondence exists in the $G_N \rightarrow 0$ limit and whether it is related to DSSYK holography [25] or not.

A Analytic continuation to euclidean signature

One representation of the de Sitter spacetime metric is

$$-X_1^2 + \sum_{i=2}^{D+1} X_i^2 = R^2 \quad (\text{A.1})$$

Analytic continuation to Euclidean signature via $X_{1,E} = iX_1$ yields

$$X_{1,E}^2 + \sum_{i=2}^{D+1} X_i^2 = R^2 \quad (\text{A.2})$$

describing the geometry of an S^D sphere with symmetry group $SO(D+1)$. Considering the correlation function on the sphere S^D , we have

$$\langle \phi(X_{1,E} \vec{Z}) \phi(X'_{1,E} \vec{Z}') \dots \rangle_{S^D} \quad (\text{A.3})$$

and analytically continuing back to Lorentzian signature yields the expectation value of the product of the indicated operators in the invariant state $|\Phi_{\text{dS}}\rangle$

$$\langle \Phi_{\text{dS}} | \phi(X_1 \vec{Z}) \phi(X'_1 \vec{Z}') \dots | \Phi_{\text{dS}} \rangle \quad (\text{A.4})$$

However, this alone does not fully justify treating the invariant state Φ_{dS} as a quantum state with the given correlation functions; we also require positivity $\langle \Phi_{\text{dS}} | \mathcal{O}^\dagger \mathcal{O} | \Phi_{\text{dS}} \rangle \geq 0$ for any operator $\mathcal{O}$. To show that Φ_{dS} is indeed a quantum state, we use the relation between Lorentzian and Euclidean signatures $X_{1,E} = iX_1$, which implies that $X_1 = 0$ corresponds to $X_{1,E} = 0$. Thus, a Lorentzian hypersurface at $X_1 = 0$ intersects a Euclidean hypersurface at $X_{1,E} = 0$. One of the most useful ways to define a quantum state on a hypersurface $\mathcal{S}$ in Euclidean signature is via a path integral over a manifold $\mathcal{W}$ whose boundary is the hypersurface $\mathcal{S}$ (see the figure below). We denote this quantum state by $\Psi_{\mathcal{W}}(\phi_{\mathcal{S}})$, where $\phi_{\mathcal{S}}$ represents an arbitrary quantum field on $\mathcal{S}$, and by definition

$$\Psi_{\mathcal{W}}(\phi_{\mathcal{S}}) = \int_{\phi_{\mathcal{W}}|_{\mathcal{S}} = \phi_{\mathcal{S}}} D\phi_{\mathcal{W}} e^{-I[\phi_{\mathcal{W}}]} = \text{diagram}$$


where $\phi_{\mathcal{W}}$ is the quantum field on the manifold $\mathcal{W}$ and $I[\phi_{\mathcal{W}}]$ is the action.

Now we compute the expectation value for the invariant state Φ_{dS} . In Euclidean signature, the stated expectation value can be regarded as a correlation function on an S^D sphere, but when all operators are placed on the equator, no analytic continuation is needed since the equator lies at $X_1 = 0$. Thus, the expectation value equals the Euclidean path integral over an S^D sphere with operators inserted on the equator.

The same calculation can be performed for the state $\Psi_{\mathcal{W}}$. Take the state $|\Psi_{\mathcal{W}}\rangle$ as the path integral over the lower hemisphere $X_1 \leq 0$, and similarly $\langle\Psi_{\mathcal{W}}|$ over the upper hemisphere $X_1 \geq 0$. Then $\langle\Psi_{\mathcal{W}}|\Psi_{\mathcal{W}}\rangle$ corresponds to gluing the two hemispheres together, and inserting operators on the hypersurface $\mathcal{S}$ yields

$$\langle\Psi_{\mathcal{W}}|O_1(x_1)O_2(x_2)\dots|\Psi_{\mathcal{W}}\rangle \quad (\text{A.5})$$

Thus, this gluing produces precisely the Euclidean path integral over the S^D sphere with operators on the equator, which is the expectation value in the invariant state Φ_{dS} :

$$\langle\Phi_{\text{dS}}|O_1(x_1)O_2(x_2)\dots|\Phi_{\text{dS}}\rangle = \langle\Psi_{\mathcal{W}}|O_1(x_1)O_2(x_2)\dots|\Psi_{\mathcal{W}}\rangle \quad (\text{A.6})$$

Therefore,

$$|\Phi_{\text{dS}}\rangle = \frac{1}{\sqrt{Z}} |\Psi_{\mathcal{W}}\rangle = \frac{1}{\sqrt{Z}} \int_{\phi_{\mathcal{W}}|_{\mathcal{S}}=\phi_{\mathcal{S}}} D\phi_{\mathcal{W}} e^{-I[\phi_{\mathcal{W}}]} \quad (\text{A.7})$$

so $|\Phi_{\text{dS}}\rangle$ can be viewed as a quantum state.

References

- [1] G. W. Gibbons and S. W. Hawking. Cosmological event horizons, thermodynamics, and particle creation. *Phys. Rev. D*, 15:2738–2751, May 1977.
- [2] Joseph Polchinski. The phase of the sum over spheres. *Phys. Lett. B*, 219:251–257, 1989.
- [3] Juan Maldacena. Real observers solving imaginary problems, 2025.
- [4] Edward Witten. Introduction to black hole thermodynamics, 2025.
- [5] Edward Witten. Gravity and the crossed product. *Journal of High Energy Physics*, 2022(10), October 2022.
- [6] Leonard Susskind. Why do we Need Observers? Spontaneous Breaking of Time-Reversal in de Sitter Space. 12 2025.
- [7] Leonard Susskind. More About the Spontaneous Breaking of Time Reversal in de Sitter Space. 1 2026.
- [8] F. J. Murray and J. v. Neumann. On Rings of Operators. *Annals Math.*, 37(1):116–229, 1936.
- [9] Hong Liu. Lectures on entanglement, von neumann algebras, and emergence of spacetime, 2025.

- [10] M. Takesaki. *Tomita's Theory of Modular Hilbert Algebras and its Applications*. Lecture Notes in Mathematics. Springer-Verlag, 1970.
- [11] M. Takesaki. *Theory of Operator Algebras II*. Encyclopaedia of Mathematical Sciences. Springer Berlin Heidelberg, 2013.
- [12] Stephen J. Summers. Tomita-takesaki modular theory, 2005.
- [13] O. Bratteli and D. W. Robinson. *OPERATOR ALGEBRAS AND QUANTUM STATISTICAL MECHANICS. 1. C^* AND W^* ALGEBRAS, SYMMETRY GROUPS, DECOMPOSITION OF STATES*. 1979.
- [14] A. van Daele. *Continuous Crossed Products and Type III Von Neumann Algebras*. London Mathematical Society Lecture Note Series. Cambridge University Press, 1978.
- [15] Julian De Vuyst, Stefan Eccles, Philipp A. Höhn, and Josh Kirklin. Crossed products and quantum reference frames: on the observer-dependence of gravitational entropy. *Journal of High Energy Physics*, 2025(7), July 2025.
- [16] Philipp A. Höhn, Alexander R. H. Smith, and Maximilian P. E. Lock. Trinity of relational quantum dynamics. *Physical Review D*, 104(6), September 2021.
- [17] Masamichi Takesaki. Duality for crossed products and the structure of von Neumann algebras of type III. *Acta Mathematica*, 131(none):249 – 310, 1973.
- [18] S W Hawking. Black holes in general relativity. *Commun. Math. Phys.* 25: No. 2, 152-66(1972)., 12 1971.
- [19] Venkatesa Chandrasekaran, Roberto Longo, Geoff Penington, and Edward Witten. An algebra of observables for de sitter space. *Journal of High Energy Physics*, 2023(2), February 2023.
- [20] Edward Witten. A background independent algebra in quantum gravity, 2023.
- [21] Jordan Cotler and Kristan Jensen. Norm of the no-boundary state, 2025.
- [22] Dionysios Anninos, Frederik Denef, Y. T. Albert Law, and Zimo Sun. Quantum de sitter horizon entropy from quasicanonical bulk, edge, sphere and topological string partition functions. *Journal of High Energy Physics*, 2022(1), January 2022.
- [23] Yiming Chen, Douglas Stanford, Haifeng Tang, and Zhenbin Yang. On the phase of the de sitter density of states, 2025.
- [24] Edward Witten. Bras and kets in euclidean path integrals, 2025.
- [25] Vladimir Narovlansky and Herman Verlinde. Double-scaled SYK and de Sitter Holography. 5 2025.